

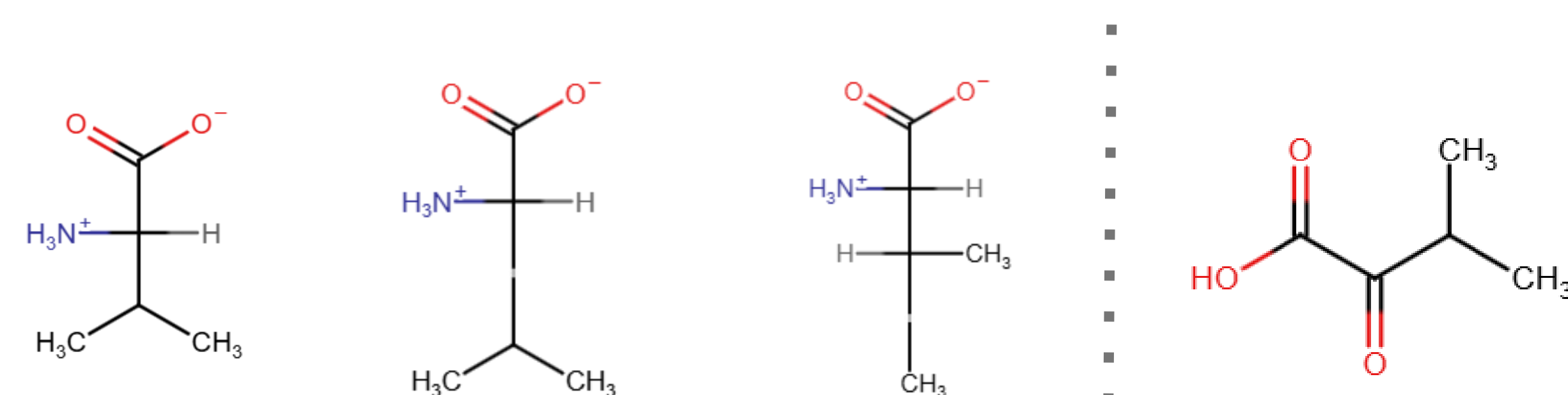
Colorimetric Assay For MSUD Detection Through Alpha Keto Acids

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INTRODUCTION

Maple syrup urine disease (MSUD) is a rare genetic disorder affecting the metabolism of branched-chain amino acids, causing the buildup of toxic byproducts, branched-chain α -ketoacids, in the body.



Left to right: three branched-chain amino acids (valine, leucine, and isoleucine), and byproduct of valine, α -ketoisovalerate

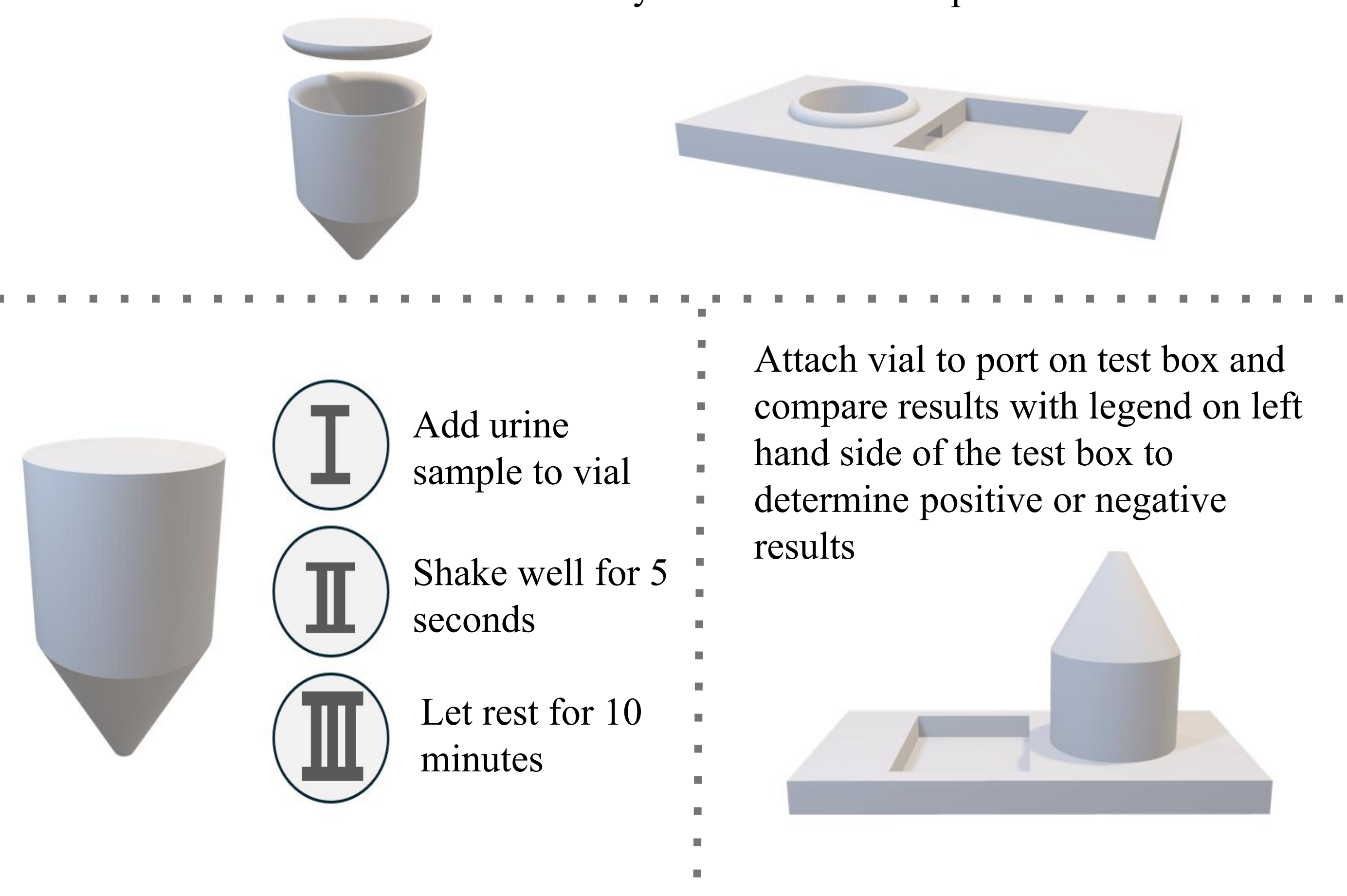
Symptoms of classical MSUD, the most severe form, include: delayed developmental milestones, lethargy, difficulty feeding, and sweet-smelling urine. Left untreated, it leads to irreversible neurological injury within 7 to 10 days after birth, and death occurs within 2 months¹.

Testing for MSUD consists of a blood or urine screening, which takes time and advanced technology. Our goal is to develop a simpler colorimetric test for the disease, targeting a byproduct of valine, α -ketoisovalerate (KIV).

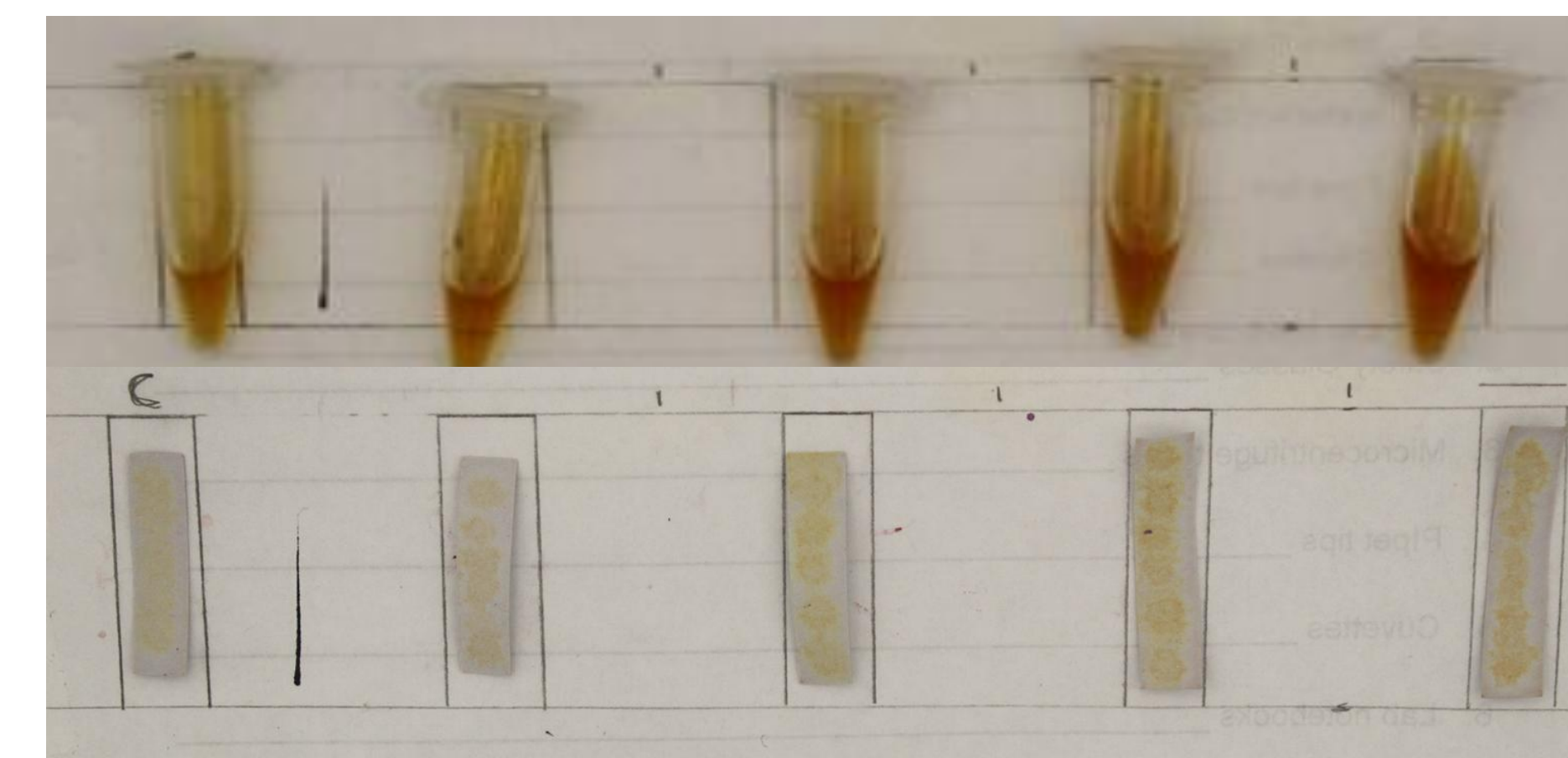
DEVICE CONCEPT

Vial with DNPH and pH of ~ 2 (regulated by Sulfuric Acid), and designed to attach to test box

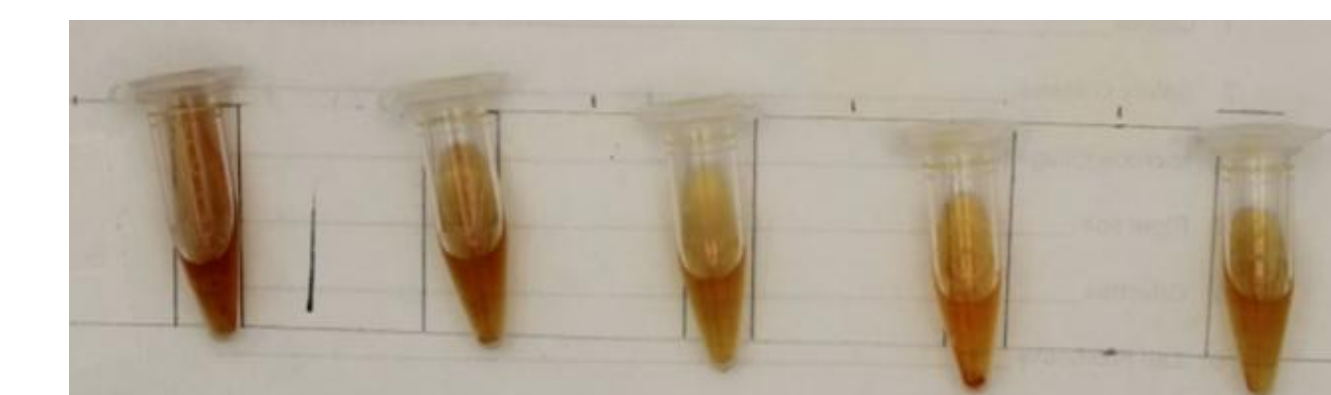
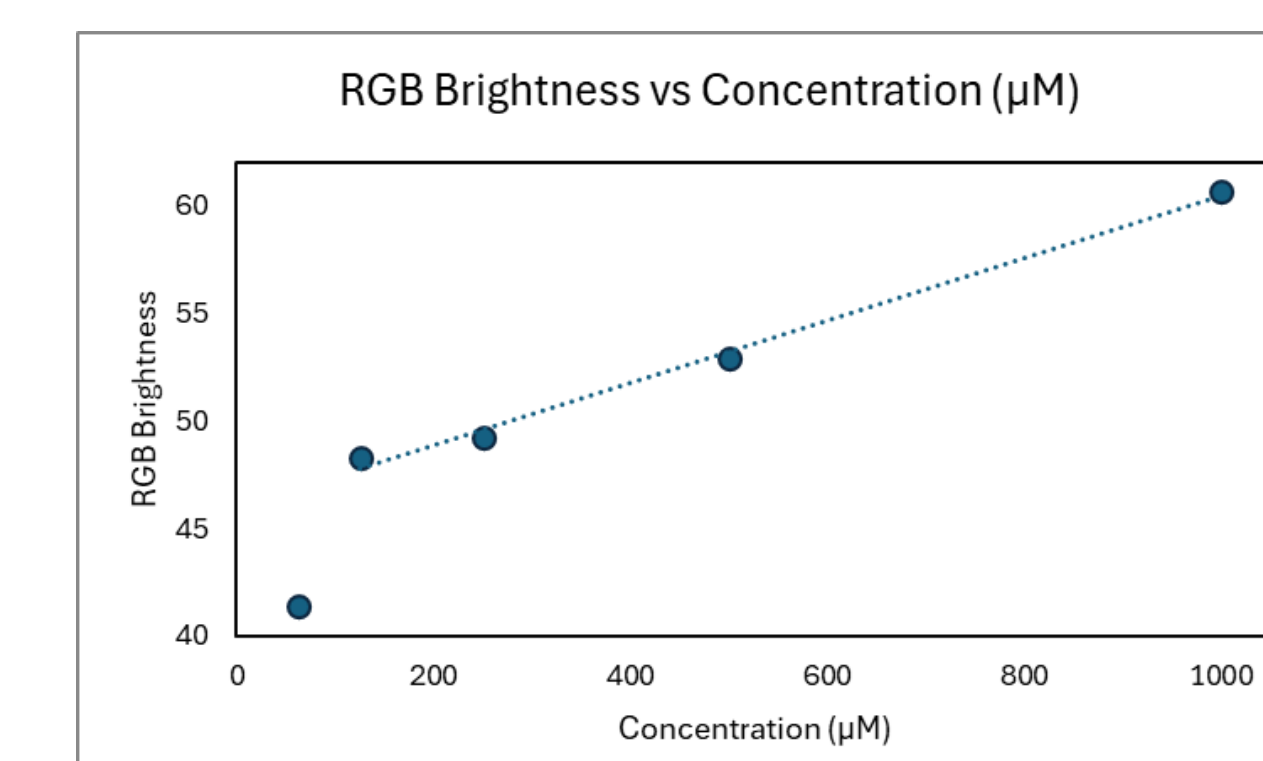
Test box with a port for the vial with the sample and a test area that contains a strip that is prepped with ninhydrin and enough sodium hydroxide to create a pH of ~ 13



RESULTS/DISCUSSION



Reactions of 1:2 serial dilutions of KIV in solution starting at 1 mM KIV (left), 1 mM DNPH, 0.1 M sulfuric acid, 8% ninhydrin, and 1 M NaOH. This reflects our current level of sensitivity, where 0.5 mM and higher concentrations would result in a positive test and 0.25 mM and lower concentrations would result in a negative test. The liquid tests have proven to be much more differentiable between different concentrations compared to paper tests, which did not show as much differentiability.

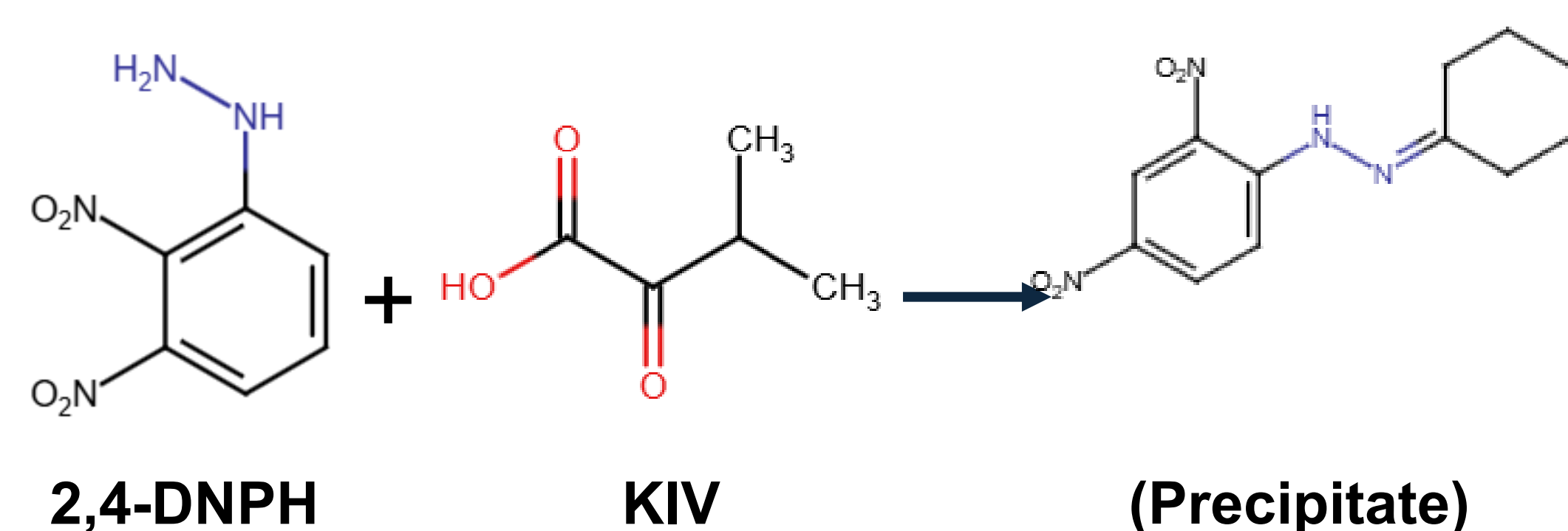
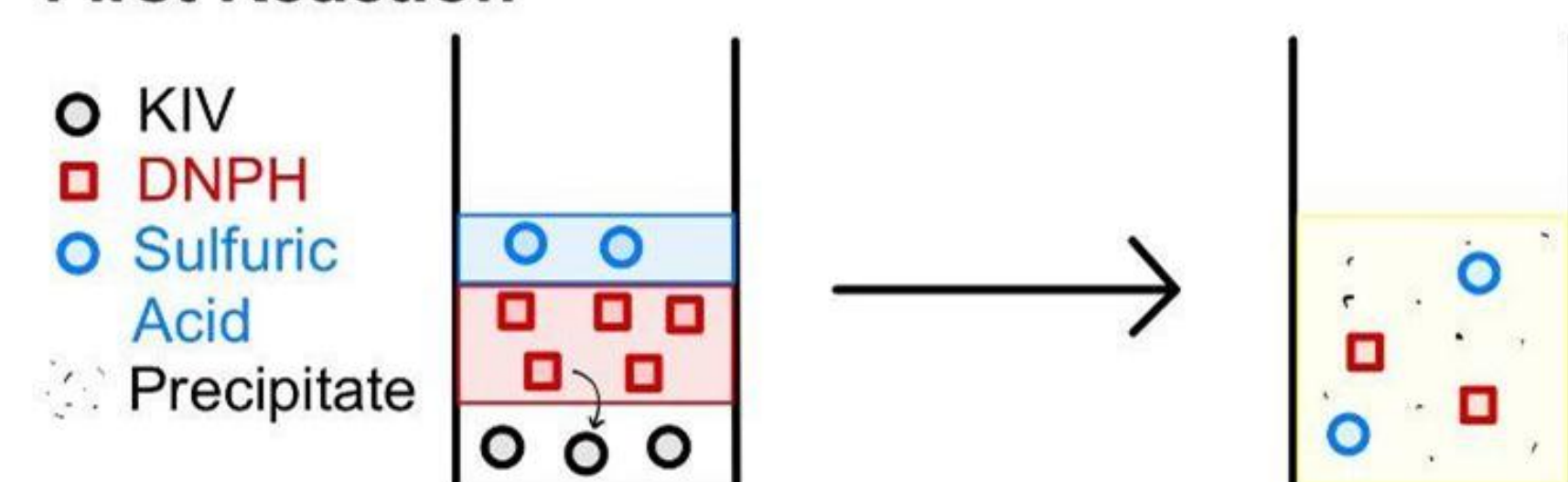


Original experiments, including 5 mM KIV (left), 5 mM DNPH, 2% ninhydrin, and 1 M NaOH, did not involve sulfuric acid, which was necessary to make Brady's Reagent rather than just 2,4-DNPH dissolved in ethanol. The exclusion of sulfuric acid gave us a color gradient that was opposite to the expected results.

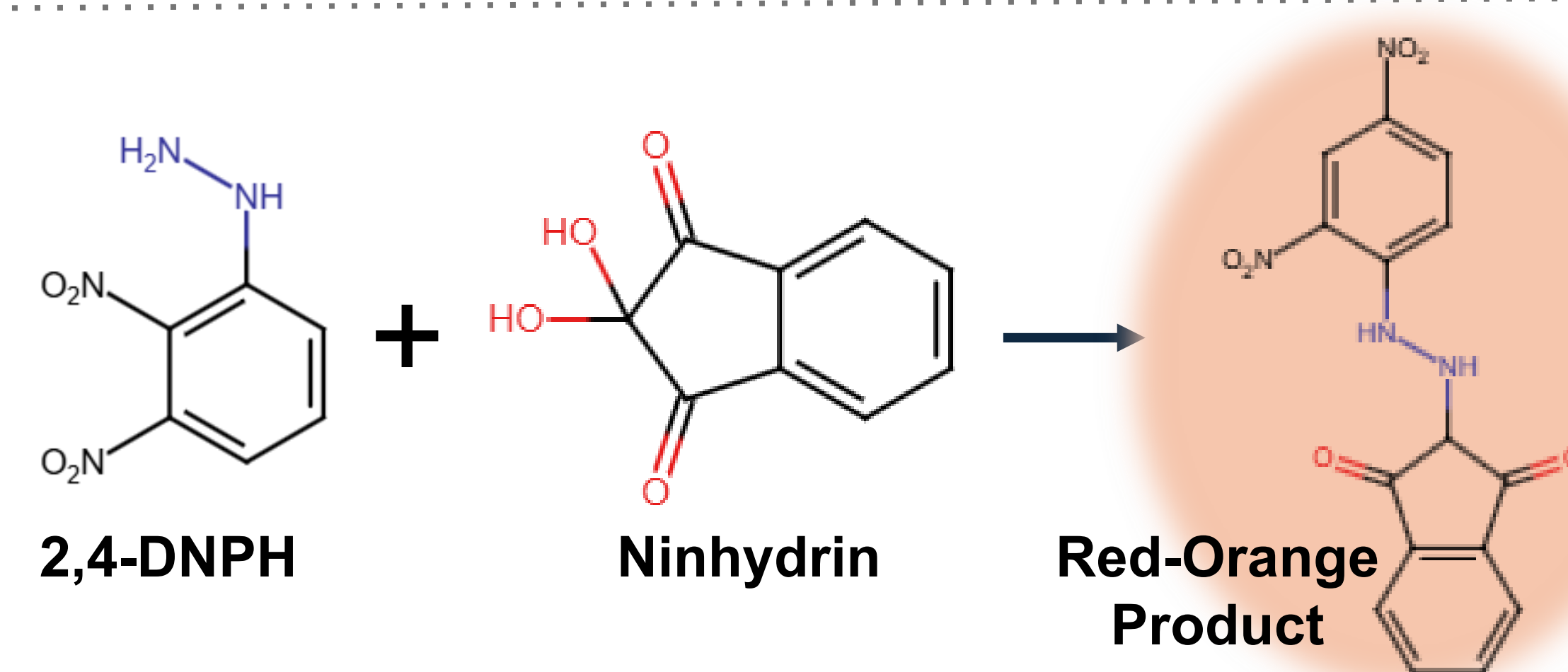
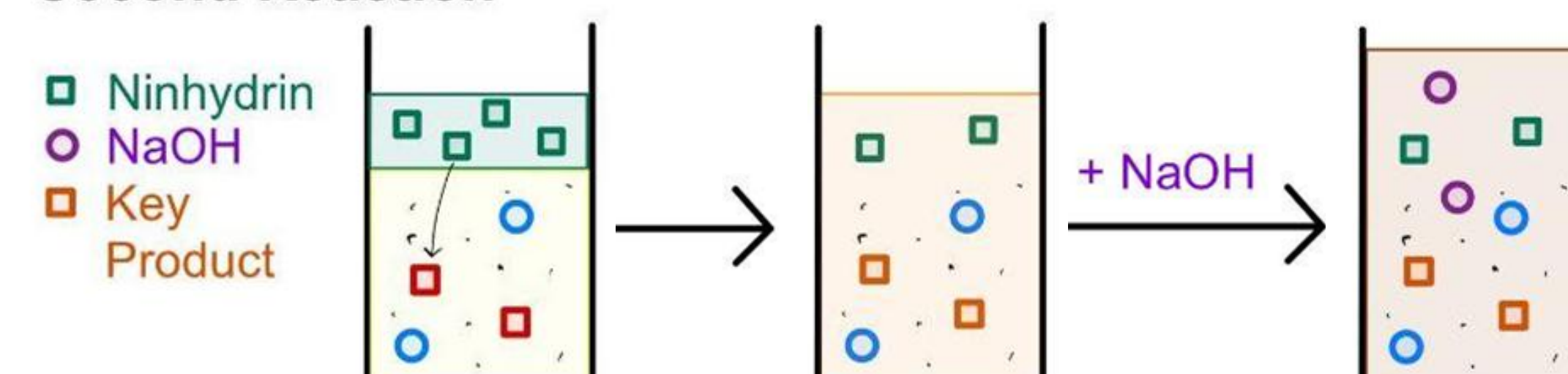
Plot of the concentrations of KIV vs. RGB brightness values from our final liquid tests. Higher RGB values indicate more brightness (higher is more yellow, lower is more red). Trendline shown for the linear portion formed by the last four data points.

METHODS

First Reaction



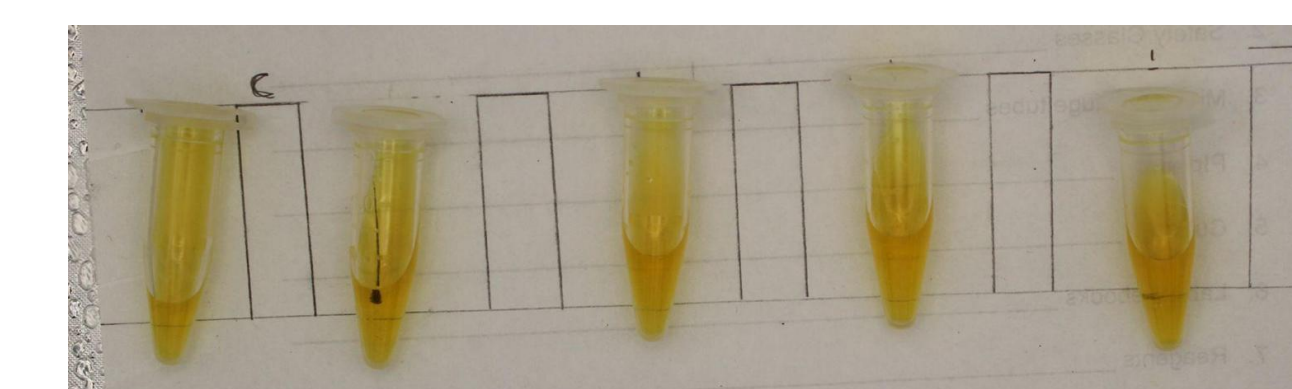
Second Reaction



Our method involves two reactions. The target, KIV, reacts with 2,4-dinitrophenylhydrazine (2,4-DNPH), forming an imperceptible precipitate. Remaining 2,4-DNPH then reacts with ninhydrin, forming a red-orange color. Lower KIV results in less 2,4-DNPH used in the first reaction, allowing for a more intense color change. Higher concentrations of KIV result in more 2,4-DNPH used, leading to less red-orange color formed.

NEXT STEPS

- Produce a more significant color change below 200 μM KIV (test of these lower concentrations with current methods shown in image)
- Develop a method to be able to show a color change (red) at a KIV concentration of 1 μM
- Further develop and finish a usable paper fluidic device



Serial dilutions of 50 μM DNPH (left) also containing 8% ninhydrin and 2 M NaOH.

REFERENCES

- Hassan SA, Gupta V. Maple Syrup Urine Disease. [Updated 2024 Mar 3]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK557773/>
- Elgendy, K.; Alaa, M.; Elmosallamy, E.; Hassan, M.; Gamal, H. Applications of Ninhydrin: Spectrophotometric Determination of Hydrazine and Hydroxylamine. *Malaysian Journal of Analytical Sciences* **2022**, 26, 855–866.
- Structures drawn with Marvin JS by Chemaxon.

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